

Instructor: Date:

Learner: Regional Branch:

Program: Center:

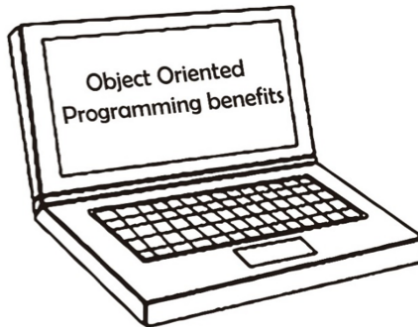
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Let's Start!

Software in an Object-Oriented Way



Learning Outcome:

Describe common simple incidents at work in a simple spoken or written situation with a supportive interlocutor. Build the macro system conceptual model following the customer requirements, through the use and interpretation of collected information, represented in class diagrams, interaction diagrams, and operation contracts.

Can Do Statements:

- I can list some benefits to use OOP.
- I can recall the ideas that are basic for OOP.
- I can talk about things to consider before writing a software.

Ask your partner:

- Think on any brand new software, what do you think are the advantages and disadvantages of that software?
- What do you think you need to create a new software?





Let's Explore!

An engineer needs to understand the benefits you get when using OOP*.

*"OOP" means Object Oriented Programming



Instructions: Read the text below.

Engineer: I need to write a new software system for the company, but I don't know if it is a good idea to use OOP.

Assistant: OOP is the best approach to writing a software. Do you remember all those benefits of the OOP?

Engineer: Yes! These benefits are always around the ideas that are basic to OOP.

Assistant: Do you know any of these ideas?

Engineer: Well, I need to review them, but I think these are: Abstraction, encapsulation, inheritance, and polymorphism.

Assistant: Are these ideas related to OOP principles?

Engineer: Well, these are also principles.

Assistant: What do you need to do with that information?

Engineer: We need to consider that information so we can develop a software model that tells us what the software must do and how it should work.

Assistant: Do we need to do this before creating our new software?

Engineer: Yes, we do. Remember that we are using fragile information, and when we create this new software, we need to make it comprehensible.

Assistant: That is why, we need to create classes by object-oriented modeling the real world, so it can be easily understood.

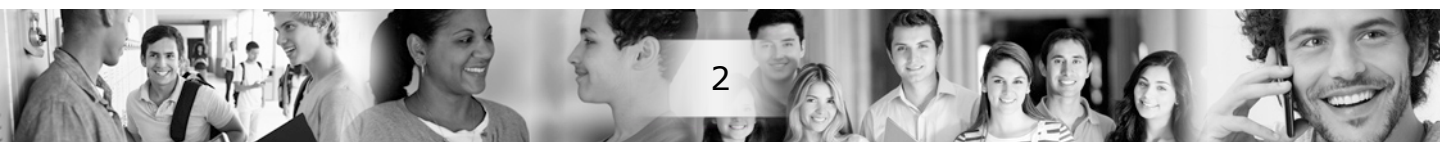
Assistant: Can you tell me the specific benefits you get when using OOP to create a software?

Engineer: Well, the benefits are better maintainability, this means that the software can be more comprehensible and less fragile; and better abstraction, which means that you can model information and behavior together.

Assistant: Do those benefits include things like good reusability?

Engineer: Well, you can also get better reusability, this means you can categorize components in general classes.

Assistant: Great! So, let's start.



**Let's Practice!**

The use of OOP brings important benefits.

Instructions: Read the text and underline all the demonstrative pronouns.

1. Which sentences are using demonstratives?
 - a. What about the benefits you get when using OOP to create a software?
 - b. These benefits are better maintainability and better abstraction.
 - c. This software can be more comprehensible and less fragile.
 - d. Better abstraction means you can model information and behavior together.
 - e. And those are the benefits you get when using OOP for software development.

**Let's Practice!**

Let's review some concepts!

Instructions: Match column A with column B.

COLUMN A

1. You need to use OOP.
2. I need to review the OOP principles.
3. You need to get better reusability,

COLUMN A

- a. I think these are: abstraction, encapsulation, inheritance and polymorphism.
- b. this means you can categorize components in general classes.
- c. to write that new software system.



Extension Activity

A poster about OOP benefits.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search more information about the benefits of using OOP.
- Choose three benefits of using OOP.
- Create a poster that explains those three benefits and promotes the use of OOP.
- Share your poster with your partner.

Self-Assessment

Yes

No

- I search information about benefits of using OOP.
- I choose 3 benefits of using OOP.
- I create a poster to promote OOP.
- I share my poster with my partner.

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Target Words

- **reusability** (noun): use of existing components within the software, this is easier to do when they are categorized in classes
e.g. Well, you can also get better reusability, this means you can categorize components in general classes.
- **class** (noun): software design that describes generalities about something
e.g. That is why, we need to create classes by object-oriented modeling the real world, so it can be easily understood.
- **inheritance** (noun): class or object you implement in the same way
e.g. Well, I need to review them, but I think these are: Abstraction, encapsulation, inheritance, and polymorphism.
- **approach** (noun): the way you deal with something
e.g. OOP is the best approach to writing a software.
- **object-oriented modelling** (noun): process for developing a software with an OOP approach
e.g. That is why, we need to create classes by object-oriented modeling the real world, so it can be easily understood.
- **software model** (noun): software prototype
e.g. We need to consider that information so we can develop a software model that tells us what the software must do and how it should work.



Learning Points

Demonstrative Pronouns in Affirmative Statements and Questions

You use **demonstrative pronouns** to point or identify specific things within a sentence.

You use ***This, That, These and Those*** as demonstrative pronouns

You use demonstrative pronouns before nouns in a sentence.

You use *This and These* to point nearby things.

You use *That and Those* to point far things.

You can use Demonstrative Pronouns in affirmative sentences or interrogative sentences. For using demonstrative pronouns within interrogative sentences, you need to use a regular grammar structure for questions and add the demonstrative pronoun right before the noun.

Demonstratives with Singular Nouns

This / That	Noun	Verb	Complement
This	software	is	empty
That	Program	Is	new
<i>Is this software empty? - Is that program new?</i>			

Demonstratives with Plural Nouns

These / Those	Noun	Verb	Complement
These	principles	Are	important
Those	programs	are	new
<i>Are these principles important? - Are those programs new?</i>			

Examples

- *Do you remember all those benefits when using OOP?*
- *Those benefits are going around these ideas that are basic for OOP.*
- *What do you need to do with that information?*





Learning Points

Verbs with Infinitives: Need to, in Affirmative Statements

You use **verb+ infinitive** in affirmative statements to express necessity of doing a certain action.

You use the following structure when writing a verb+ infinitive in affirmative statements with "need to".

Subject + Need + to + Basic form of a verb

Example:

I need to design a model

Remember to add an -s to the verb **need** when used with third person singular nouns. The verb that comes after the preposition *to*, always keeps its basic form:

Example:

She needs to learn about OOP.

Examples

- *I need to write a new software system for the company*
- *We need to consider that information and we need to develop a software model that tells us what the software has to do and how it should work.*
- *When we create this new software we need to make it comprehensible.*